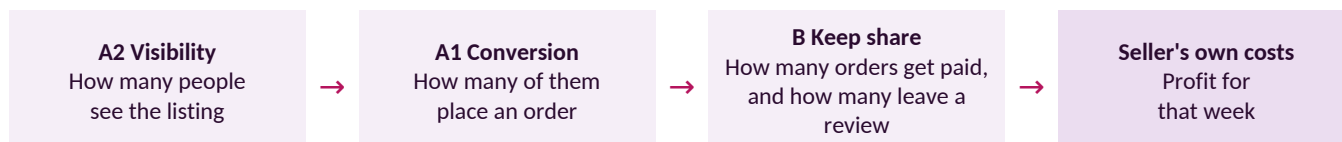


Price Confidence: how the three models work

What each model learns, how it is trained, what it outputs, and how it changes the price a seller sees. Numbers are from our demo, trained on 300 synthetic listings (150 soap, 150 t-shirt).



New sales and reviews from this week feed next week's A2 and A1. The app repeats the chain week by week, for every candidate price, over 12 weeks (26 for the launch-lower test), for a good, a typical and a weak product.

The training data and the one rule all three models follow

One row = one listing in one week: **3,909 soap rows and 3,718 t-shirt rows**, 150 listings each, launched Sep 2025 to Aug 2026 and followed to 21 Sep 2026. Inputs: **relative price** = price ÷ market median – 1; **rating shown** that week, pulled toward the typical 3.9★ when reviews are few; **reviews so far**; **recent sales** (conversion versus the category norm); launch phase; category demand; listing quality. **The rule**: every model gives each listing its own baseline (fixed effects), so effects are learnt from a listing changing over time, never from comparing good products with bad ones. A second step then puts those baselines on listing quality, so a brand-new listing gets a starting level.

A1 Conversion: of the people who see it, how many order?

Trained on: orders in the week with impressions as the exposure (Poisson regression with a log-impressions offset), so more views are never mistaken for better conversion. **Why baselines matter**: comparing different listings would say +10% price costs only 7% of orders (soap): better products charge more and still sell more.

What it learnt	Soap	T-shirt	Meaning
Price vs market	–2.40 (curve +1.36)	–3.20 (curve +1.39)	+10% price: –20% orders per impression (soap), –26% (t-shirt)
Rating shown, per star	0.49	0.65	+0.3★: +16% conversion (soap), +21% (t-shirt)
Reviews so far	0.01	0.00	About zero: reviews work through visibility (A2)
Listing quality, per SD	0.68	0.71	Better-presented listings convert more
Spread between products	0.44	0.43	A top-quarter product converts about 35% better than a typical one

How it moves the answer: the demand curve. For Seller 6's soap in launch week, conversion is 0.125% at ₹179, 0.087% at ₹209 and 0.070% at ₹229.

A2 Visibility: how many people see the listing each week?

Trained on: weekly impressions, Poisson regression with listing baselines. The earlier version without baselines said recent sales barely affect visibility (0.007); within listings the effect is 0.30. Same trap as in A1.

What it learnt	Soap	T-shirt	Meaning
Launch weeks 1–2 / 3–4	0.59 / 0.32	0.61 / 0.34	A new listing is shown 1.8× , then 1.4×, then normal
Recent sales vs category norm	0.30	0.29	Selling at twice the norm: +35% impressions next week
Reviews so far	0.10	0.07	Reviews double: +7% (soap), +5% (t-shirt)
Rating shown, per star	0.22	0.54	+0.3★: +7% (soap), +17% (t-shirt)
Category demand / sale week	1.07 / 0.25	0.97 / 0.27	Demand +10%: about +10%; sale week: +29% (soap)

How it moves the answer: Seller 6 is shown about 10,200 times in week 1 and about 5,400 by week 5. For a good product, launching at ₹129 brings about 4.6× the orders in week 2, which lifts week-5 impressions by about 56%. That is real, but the discount costs more than the extra visibility earns back.

B Keep share: of the orders, how many get paid, and how many review?

Part	How it is trained	Soap	T-shirt
RTO rate	Total RTO ÷ total orders (it did not depend on price)	10.1%	12.5%
Return rate	Logistic regression with listing baselines; rises with price above the market; better listings return less	6.3% at ₹209	21% at ₹349
Good vs weak product	Spread of return rates between products, noise removed	4.8% / 8.3%	spread 0.38
Returns resold / review rate	Plain shares from the data	0% / 14%	70% / 15%

We force the price effect on returns to be zero or positive: left free, the model learnt that pricier listings get fewer returns, the same quality trap. **How it moves the answer**: $(1 - 10.1\%) \times (1 - 6\%) = 84.5\%$ of Seller 6's parcels get paid; costs are divided by that, which gives his ₹145 floor. The review rate turns orders into next week's reviews.

All three together: one simulated week

Step	Model	Seller 6, typical product, ₹209, week 1
1. How many see it	A2	About 10,200 impressions (launch boost, no reviews yet)
2. How many order	A1	0.087% conversion: about 8.9 orders placed
3. How many pay	B	84.5% paid: about 7.5 paid orders; about 1.1 new reviews
4. Profit	Seller's costs	₹51 per dispatched order × 8.9 = about ₹455 for the week
5. Next week	Loop	Sales and reviews feed A2 and A1; the launch boost fades; repeat

Launch price	Orders placed (paid), 12 weeks	Profit, 12 weeks	Why
₹179	107 (90)	₹2,973	More orders, but each earns too little
₹209 (chosen)	69 (58)	₹3,537	Within 5% of the best, with more orders, so reviews come sooner. 58 paid × ₹61 = ₹3,537
₹229	54 (46)	₹3,627	Highest profit, but fewer orders and slower reviews

Checked against the simulator that generated the data

We ran Seller 6's soap through the simulator's own buyer rules, 300 times per plan, and compared the change in 26-week profit against simply starting at ₹209.

Plan	Good product: model / simulator	Typical: model / simulator	Weak: model / simulator
₹129 for 4 weeks (below the ₹145 floor)	-₹2,626 / -₹779	-₹2,249 / -₹1,308	-₹1,688 / -₹1,698
₹169 for 4 weeks (above the floor)	+₹72 / +₹1,690	-₹241 / +₹531	-₹277 / -₹15
Reviews after 4 weeks at ₹129	26 / 35	14 / 20	8 / 11

What it shows: both agree that launching below the floor loses money for every kind of product. The models are more cautious than the simulator about discounts that stay above the floor: sellers in the data cut prices when sales are slow, which blurs the price effect. Only a randomised price test removes that, so the pilot includes one.

What is “listing quality”?

A 0–100 score for **how well the product is presented**: clear photos, a plain well-lit background, a full description, a size chart or ingredient list. It is not product quality, which buyers learn after delivery through ratings and returns. In the app the seller ticks 4 boxes, each worth 7.5 points (score 50 to 80). **One extra box (soap):** conversion about +47%, impressions +5%, returns about 11% lower. The conversion effect is large because the score also stands in for product quality the model can't see; a real version would compute it from Meesho's catalog data and test it.

Which app output comes from which model, and how good it is

What the seller sees	Comes from
Floor price and the bill for one paid order	B only, plus the seller's own costs
Start price and good range	A1 + A2 + B, simulated for 12 weeks at every candidate price
Launch lower: reviews bought, cost per review, payback	A2's sales-to-visibility loop, A1 and B, over 26 weeks
“At 4.1★ you can charge about 8% more”	Rating and price effects from A1 and A2 (demo data)

Backtest: hide 20% of listings, retrain, predict their 8-week orders: typical error **31% (soap)** and **41% (t-shirt)**, against 153% and 87% for a simple average. **Limits:** synthetic data, two categories, simple regressions chosen so every number can be explained. **For real:** train on Meesho's order, return and review history, retrain regularly, and A/B test every price change.